

Xilin Zhu

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RESEARCH INTERESTS

I am interested in Computer Graphics and Generative Models. Currently, my research focuses on exploring how generative models could facilitate 3D reconstruction tasks and image intrinsic recovery.

EDUCATION

Rochester Institute of Technology, Rochester, NY Aug. 2026 - Present
PhD in Computing and Information Sciences

Tianjin Normal University, Tianjin, China Sept. 2018 - June. 2024
B.Eng. in Software Engineering
GPA: 3.6/4.0

WORK EXPERIENCE

Rochester Institute of Technology, Remote Apr. 2025 - July. 2026
Research Intern, DISCO Research Group
Working on diffusion model-based inverse rendering project.
Advisor: Prof. Yiqin Zhao

Nanjing University, Suzhou, China Oct. 2024 - Sept. 2025
Research Assistant, School of Intelligence Science and Technology
Working on projects about Appearance Modeling.
Advisor: Prof. Beibei Wang

PROJECTS

Microfacet BRDF Implementation in Mitsuba Renderer Sept. 2024 - Oct. 2024
Implemented a physically-based Microfacet BRDF with Beckmann distribution in Mitsuba renderer, including core functions (eval, pdf, sample) and performance optimization through importance sampling.

Ray Tracer from Scratch in C++ Nov. 2023 - Apr. 2024
Implemented a ray tracer from the ground up using C++.
Developed material models for diffuse surfaces.
Implemented path tracing.

Study on Weight Analysis of Ray Tracing Algorithm Nov. 2023 - Apr. 2024
Based on Combining Sampling (Undergraduate Thesis)
Conducted a study on mixed sampling strategies, specifically combining light source sampling and BRDF importance sampling.

SKILLS

Programming	C/C++, Python
Computer Graphics	Ray Tracing, 3D GS, NeRF
Generative Models	Diffusion Model, Flow Model
Developer Tools	VS/VS Code, Blender, LaTeX
Frameworks	PyTorch, Mitsuba Renderer
Language	English(TOEFL 103/120), Mandarin

AWARDS

Outstanding Undergraduate Thesis	May. 2024
Third-grade scholarship in the academic year of 2022 - 2023	Sept. 2023
College grade merit student in the academic year of 2021 - 2022	Sept. 2022
Second-grade scholarship in the academic year of 2019 - 2020	Sept. 2020
2020 "Challenge Cup" silver medal	Aug. 2020